Ron Bibb
Independent Researcher
Lilburn, Georgia, USA
ronbibb@gmail.com

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Editors
General Relativity and Gravitation

Dear Editors,

Please consider the manuscript "A Strict Surface-Energy Obstruction for Timelike Kantowski–Sachs–Schwarzschild Junctions" for publication in *General Relativity and Gravitation*. We prove that a homogeneous Kantowski–Sachs region cannot be joined to the retained ordinary asymptotically flat Schwarzschild exterior across a timelike shell without negative surface energy density, for every compatible finite-rapidity embedding and either explicitly retained Kantowski–Sachs interval. For an exact static comoving family, the null energy condition additionally fails precisely outside the Schwarzschild photon sphere, with saturation at $R_0 = 3m$, and an explicit Nariai product region supplies a local geometric witness.

The significance of the result is that it separates an invariant obstruction caused by the Kantowski–Sachs product geometry from coordinate, trajectory, and orientation artifacts. It also supplies a classical boundary diagnostic for proposed black-hole-to-cosmology completions and for effective black-hole interiors formulated in Kantowski–Sachs variables. This combination of exact junction analysis, global branch bookkeeping, and energy-condition classification is well aligned with the scope of *General Relativity and Gravitation*.

The manuscript is original, is not under consideration elsewhere, and has been prepared independently without institutional or grant support. The manuscript source, figure-generation scripts, and pinned verification suite are publicly available at github.com/RonBibb/junction-ec-paper.

Subject to the journal's conflict-of-interest checks, potential referees with relevant expertise include João L. Rosa, Sebastian Carloni, S. H. Mazharimousavi, José M. M. Senovilla, and S. Khakshournia.

Sincerely,

Ron Bibb
Independent Researcher
Lilburn, Georgia, USA
ronbibb@gmail.com
ORCID: 0009-0004-1153-2464